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LAB_02

Task 1:

Creating a replica and making sure it received the data:

```
C:\Users\ZOUNY\distributed-lab>docker exec -it redis-master redis-cli set lab_test "HelloDistributed"
OK

C:\Users\ZOUNY\distributed-lab>docker exec -it redis-replica redis-cli get lab_test
"HelloDistributed"
```

Stopping and starting the replica to observe the change in responses:

```
C:\Users\ZOUNY\distributed-lab>docker exec -it redis-master redis-cli set lab_test "HelloDistributed"
OK

C:\Users\ZOUNY\distributed-lab>docker exec -it redis-replica redis-cli get lab_test
"HelloDistributed"

C:\Users\ZOUNY\distributed-lab>docker compose stop redis-replica
[+] Stopping 1/1
  Container redis-replica  Stopped                                0.4s

C:\Users\ZOUNY\distributed-lab>docker exec -it redis-master redis-cli set lab_test "UpdatedValue"
OK

C:\Users\ZOUNY\distributed-lab>docker exec -it redis-master redis-cli set test_key "updated_during_partition"
OK

C:\Users\ZOUNY\distributed-lab>docker exec -it redis-replica redis-cli get test_key
Error response from daemon: container 86bc9c554b1ab9b61c950d047e23dc1a07e18fd97105f45616d93b97a6394ccd is not running

C:\Users\ZOUNY\distributed-lab>docker exec -it redis-replica redis-cli get test_key
Error response from daemon: container 86bc9c554b1ab9b61c950d047e23dc1a07e18fd97105f45616d93b97a6394ccd is not running

C:\Users\ZOUNY\distributed-lab>docker compose start redis-replica
[+] Running 1/1
  Container redis-replica  Started

C:\Users\ZOUNY\distributed-lab>docker exec -it redis-replica redis-cli get test_key
"updated_during_partition"
```

Using the docker compose up command to create a container with redis-master and redis replica and an etcd-node:

```
C:\Users\ZOUNY\distributed-lab>docker compose up -d
[+] Running 4/4
  Network distributed-lab_default  Created                                0.1s
  Container etcd-node             Started                             1.3s
  Container redis-master          Started                             1.3s
  Container redis-replica         Started                             1.3s
```

TASK 2:

Observing leader node using etcdctl endpoint status and stopping the etcd container and observe leader re-election

(i only had a single etcd node so when i stopped it there was no other nodes to take its place in the re-election so my etcd command failed)

```
C:\Users\ZOUNY\distributed-lab>docker exec -it etcd-node etcdctl put foo bar
OK
C:\Users\ZOUNY\distributed-lab>docker exec -it etcd-node etcdctl get foo
foo
bar
C:\Users\ZOUNY\distributed-lab>docker exec -it etcd-node etcdctl endpoint status --write-out=table
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| ENDPOINT | ID | VERSION | DB SIZE | IS LEADER | IS LEARNER | RAFT TERM | RAFT INDEX | RAFT APPLIED INDEX | ERRORS |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 127.0.0.1:2379 | 8e9e05c52164694d | 3.5.0 | 20 kB | true | false | 2 | 5 | 5 | |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
C:\Users\ZOUNY\distributed-lab>docker compose stop etcd
[+] Stopping 1/1
  Container etcd-node Stopped
C:\Users\ZOUNY\distributed-lab>docker exec -it etcd-node etcdctl endpoint status
Error response from daemon: container 204ec4f2937e322a456cc91a41657205d3cbd58a342b40914250731bd5a3dda2 is not running
C:\Users\ZOUNY\distributed-lab>
```

Cleanup

```
C:\Users\ZOUNY\distributed-lab>docker-compose down
[+] Running 4/4
  Container etcd-node Removed
  Container redis-master Removed
  Container redis-replica Removed
  Network distributed-lab_default Removed
C:\Users\ZOUNY\distributed-lab>
```